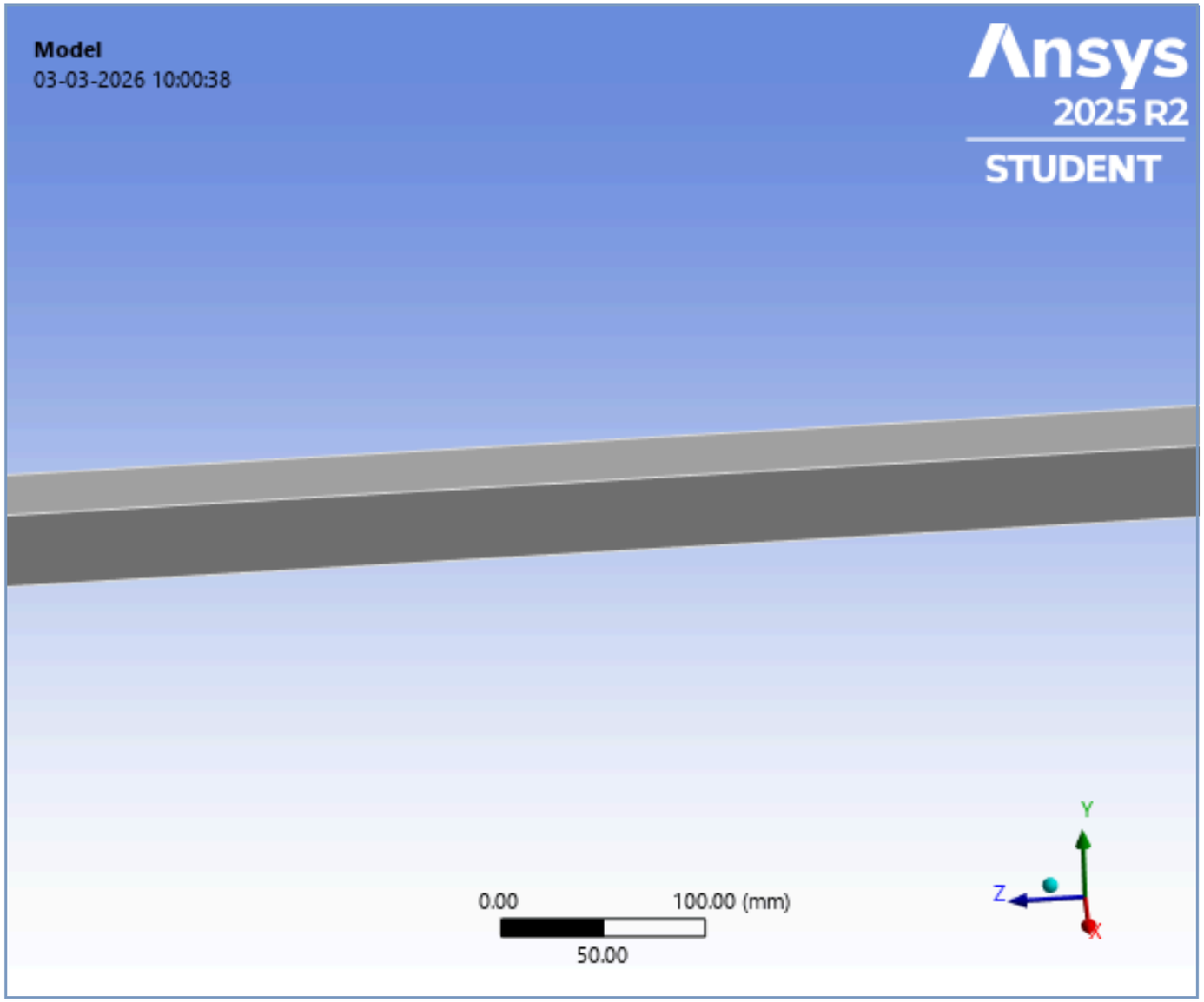
From:
To:
Date: 3/3/2026 5:28:08 PM
Subject:

Project*

First Saved	Tuesday, March 3, 2026
Last Saved	Tuesday, March 3, 2026
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No


Model
03-03-2026 10:00:38
Ansys
2025 R2
STUDENT
0.00
100.00 (mm)
50.00
Y
Z
X

Contents

- Units
- Model (A4)
  - Geometry Imports
    - Geometry Import (A3)
  - Geometry
    - Solid
  - Construction Geometry
    - Path
  - Materials
  - Coordinate Systems
  - Mesh
  - Static Structural (A5)
    - Analysis Settings
    - Loads
    - Solution (A6)
      - Solution Information
      - Results
- Material Data
  - Structural Steel

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	Geometry Imports
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	Geometry Import (A3)
State	Solved
Definition	
Source	C:\Users\dheeraj\AppData\Local\Temp\WB_dheeraj_21960_2\wbnew_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Advanced Geometry Options	
Compare Parts On Update	No
Analysis Type	3-D

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\dheeraj\AppData\Local\Temp\WB_dheeraj_21960_2\wbnew_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	40. mm
Length Y	40. mm
Length Z	1000. mm
Properties	
Volume	1.6e+006 mm^3
Mass	12.56 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	61425
Elements	12800
Mesh Metric	None
Update Options	

Assign Default Material	No
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	Solid
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes

Thermal Strain Effects	Yes
Bounding Box	
Length X	40. mm
Length Y	40. mm
Length Z	1000. mm
Properties	
Volume	1.6e+006 mm³
Mass	12.56 kg
Centroid X	-9.1154e-002 mm
Centroid Y	-0.1506 mm
Centroid Z	500. mm
Moment of Inertia Ip1	1.0483e+006 kg·mm²
Moment of Inertia Ip2	1.0483e+006 kg·mm²
Moment of Inertia Ip3	3349.3 kg·mm²
Statistics	
Nodes	61425
Elements	12800
Mesh Metric	None

TABLE 6
Model (A4) > Construction Geometry

Object Name	Construction Geometry
State	Fully Defined
Display	
Show Mesh	No

TABLE 7
Model (A4) > Construction Geometry > Paths

Object Name	Path
State	Fully Defined
Definition	
Path Type	Two Points
Path Coordinate System	Global Coordinate System
Number of Sampling Points	99.
Suppressed	No
Start	
Coordinate System	Global Coordinate System
Start X Coordinate	-9.1154e-002 mm
Start Y Coordinate	19.849 mm
Start Z Coordinate	1000. mm
Location	Defined
End	

Coordinate System	Global Coordinate System
End X Coordinate	-9.1154e-002 mm
End Y Coordinate	19.849 mm
End Z Coordinate	0. mm
Location	Defined

TABLE 8
Model (A4) > Materials

Object Name	Materials
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

Coordinate Systems

TABLE 9
Model (A4) > Coordinate Systems > Coordinate System

Object Name	Global Coordinate System
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Mesh

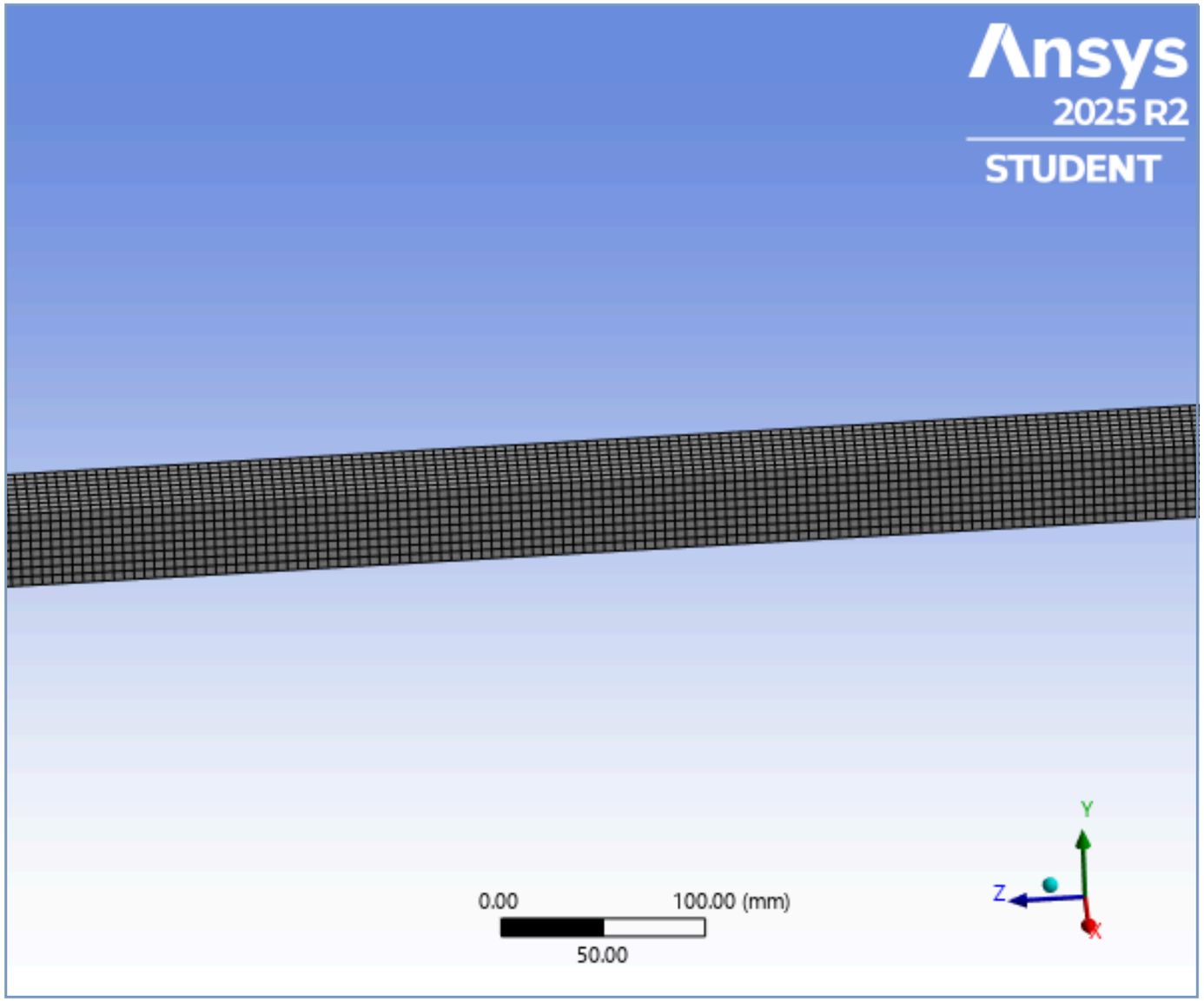
TABLE 10
Model (A4) > Mesh

Object Name	Mesh
State	Solved

Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	5.0 mm
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	1001.6 mm
Average Surface Area	27200 mm²
Minimum Edge Length	40.0 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No

Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	61425
Elements	12800
Show Detailed Statistics	No

FIGURE 1
Model (A4) > Mesh > Figure



Static Structural (A5)

TABLE 11
Model (A4) > Analysis

Object Name	Static Structural (A5)
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 12
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	Analysis Settings
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled

Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\dheeraj\AppData\Local\Temp\WB_dheeraj_21960_2\wbnew_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	nmm

TABLE 13
Model (A4) > Static Structural (A5) > Loads

Object Name	Fixed Support	Displacement	Force
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	1 Edge		1 Face
Definition			
Type	Fixed Support	Displacement	Force

Suppressed	No		
Define By		Components	Vector
Coordinate System		Global Coordinate System	
X Component		0. mm (ramped)	
Y Component		0. mm (ramped)	
Z Component		Free	
Applied By			Surface Effect
Magnitude			5000. N (ramped)
Direction			Defined

FIGURE 2
Model (A4) > Static Structural (A5) > Displacement

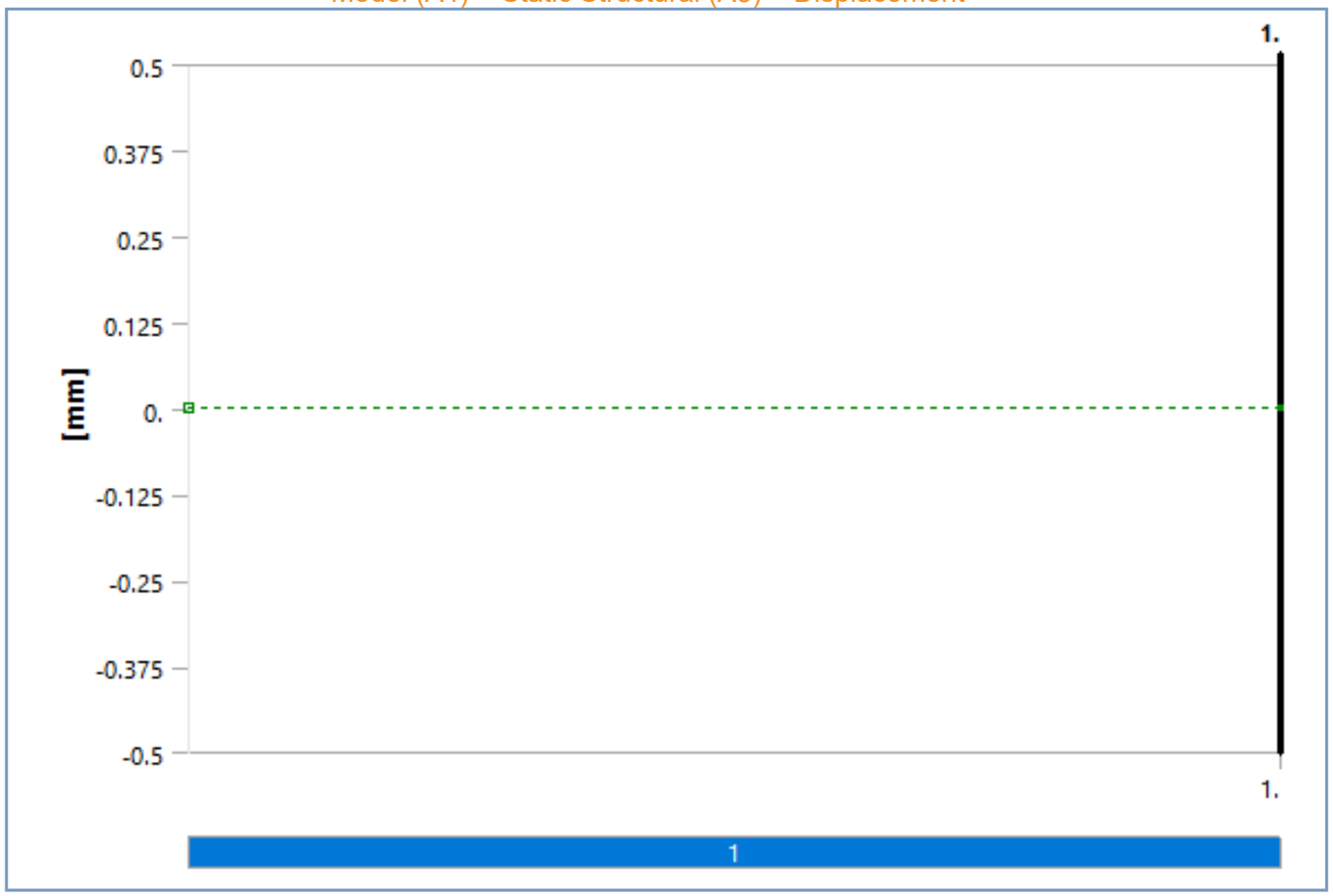
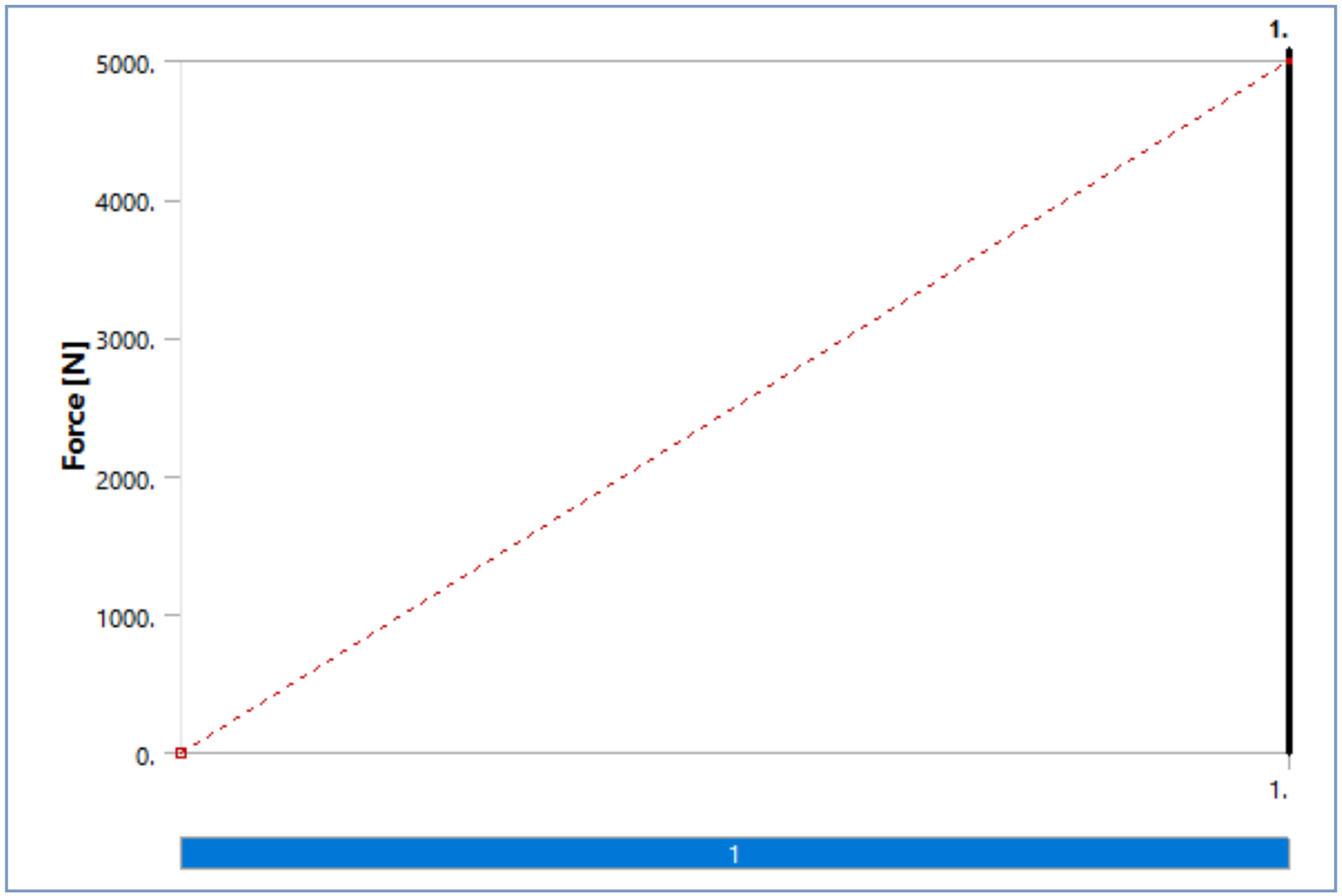


FIGURE 3
Model (A4) > Static Structural (A5) > Force



Solution (A6)

TABLE 14
Model (A4) > Static Structural (A5) > Solution

Object Name	Solution (A6)
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	3.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	20. s
MAPDL Memory Used	469. MB
MAPDL Result File Size	15. MB
Post Processing	
Beam Section Results	No

On Demand Stress/Strain	No

TABLE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	Solution Information
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 16
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Directional Deformation	Equivalent Stress
State	Solved	
Scope		
Scoping Method	Path	
Path	Path	
Geometry	All Bodies	
Definition		
Type	Directional Deformation	Equivalent (von-Mises) Stress
Orientation	Y Axis	
By	Time	
Display Time	Last	
Coordinate System	Global Coordinate System	
Calculate Time History	Yes	
Suppressed	No	
Results		
Minimum	-1.4594 mm	0.51956 MPa
Maximum	-2.3134e-003 mm	58.555 MPa
Average	-0.92559 mm	38.583 MPa
Minimum Occurs On	Solid	

Maximum Occurs On	Solid	
Graph Controls		
X-Axis	S	
Information		
Time	1. s	
Load Step	1	
Substep	1	
Iteration Number	1	
Integration Point Results		
Display Option		Averaged
Average Across Bodies		No

FIGURE 4
Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation

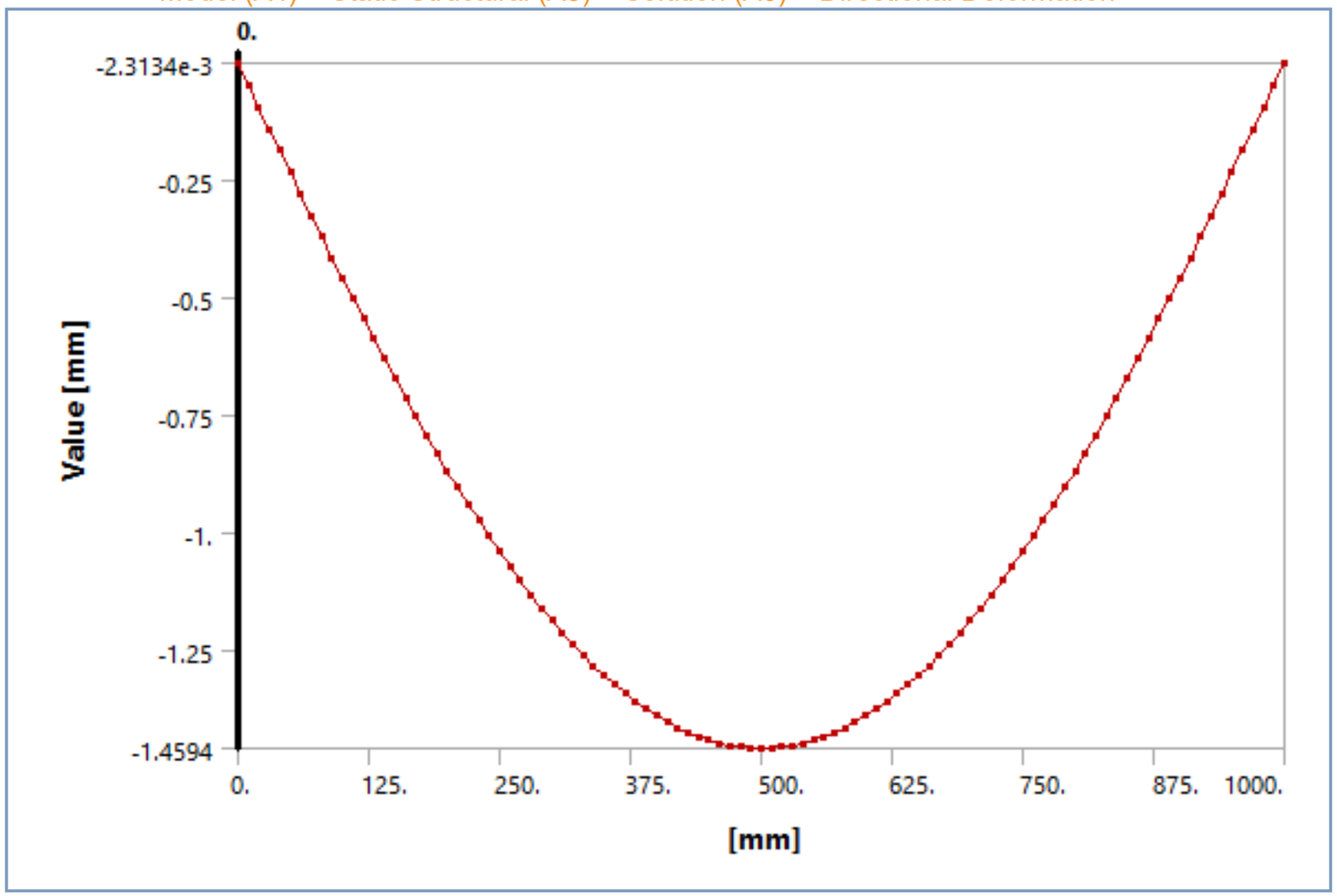


TABLE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation

Length [mm]	Value [mm]

0.	-2.3134e-003
10.	-4.8798e-002
20.	-9.5328e-002
30.	-0.14181
40.	-0.18813
50.	-0.23424
60.	-0.28008
70.	-0.32559
80.	-0.37074
90.	-0.41548
100.	-0.45976
110.	-0.50353
120.	-0.54675
130.	-0.58938
140.	-0.63138
150.	-0.6727
160.	-0.7133
170.	-0.75315
180.	-0.79221
190.	-0.83045
200.	-0.86782
210.	-0.9043
220.	-0.93985
230.	-0.97443
240.	-1.008
250.	-1.0406
260.	-1.0721
270.	-1.1026
280.	-1.1319
290.	-1.1601
300.	-1.1872
310.	-1.2131
320.	-1.2378
330.	-1.2613
340.	-1.2835
350.	-1.3045
360.	-1.3242
370.	-1.3426
380.	-1.3597
390.	-1.3755
400.	-1.39

410.	-1.4031
420.	-1.4149
430.	-1.4253
440.	-1.4343
450.	-1.442
460.	-1.4483
470.	-1.4531
480.	-1.4566
490.	-1.4587
500.	-1.4594
510.	-1.4587
520.	-1.4566
530.	-1.4531
540.	-1.4483
550.	-1.442
560.	-1.4343
570.	-1.4253
580.	-1.4149
590.	-1.4031
600.	-1.39
610.	-1.3755
620.	-1.3597
630.	-1.3426
640.	-1.3242
650.	-1.3045
660.	-1.2835
670.	-1.2613
680.	-1.2378
690.	-1.2131
700.	-1.1872
710.	-1.1601
720.	-1.1319
730.	-1.1026
740.	-1.0721
750.	-1.0406
760.	-1.008
770.	-0.97443
780.	-0.93985
790.	-0.9043
800.	-0.86782
810.	-0.83045

820.	-0.79222
830.	-0.75315
840.	-0.7133
850.	-0.6727
860.	-0.63138
870.	-0.58938
880.	-0.54675
890.	-0.50353
900.	-0.45976
910.	-0.41548
920.	-0.37075
930.	-0.32559
940.	-0.28008
950.	-0.23424
960.	-0.18813
970.	-0.14181
980.	-9.5329e-002
990.	-4.8798e-002
1000.	-2.3136e-003

FIGURE 5
Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation > Figure

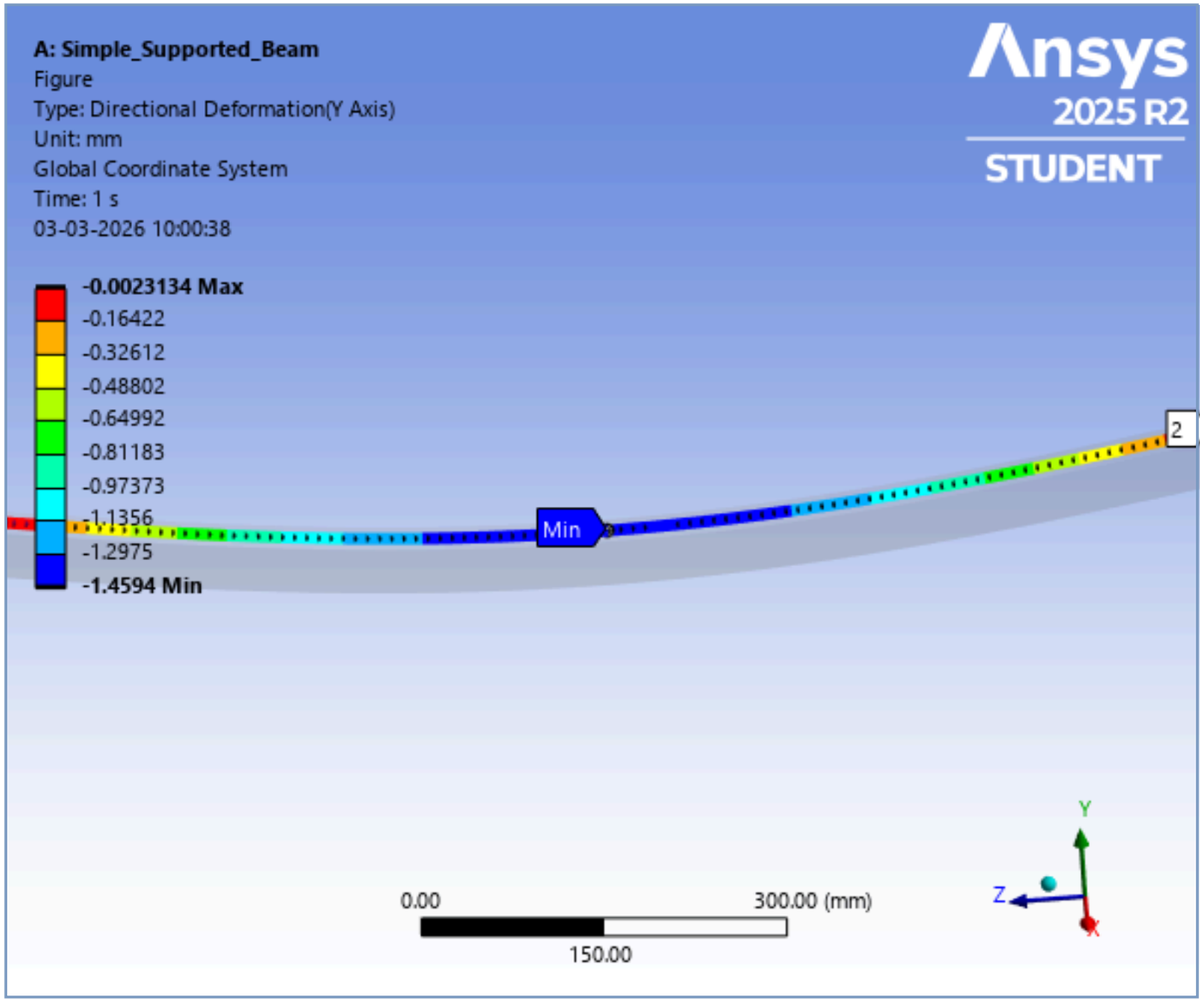


FIGURE 6
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

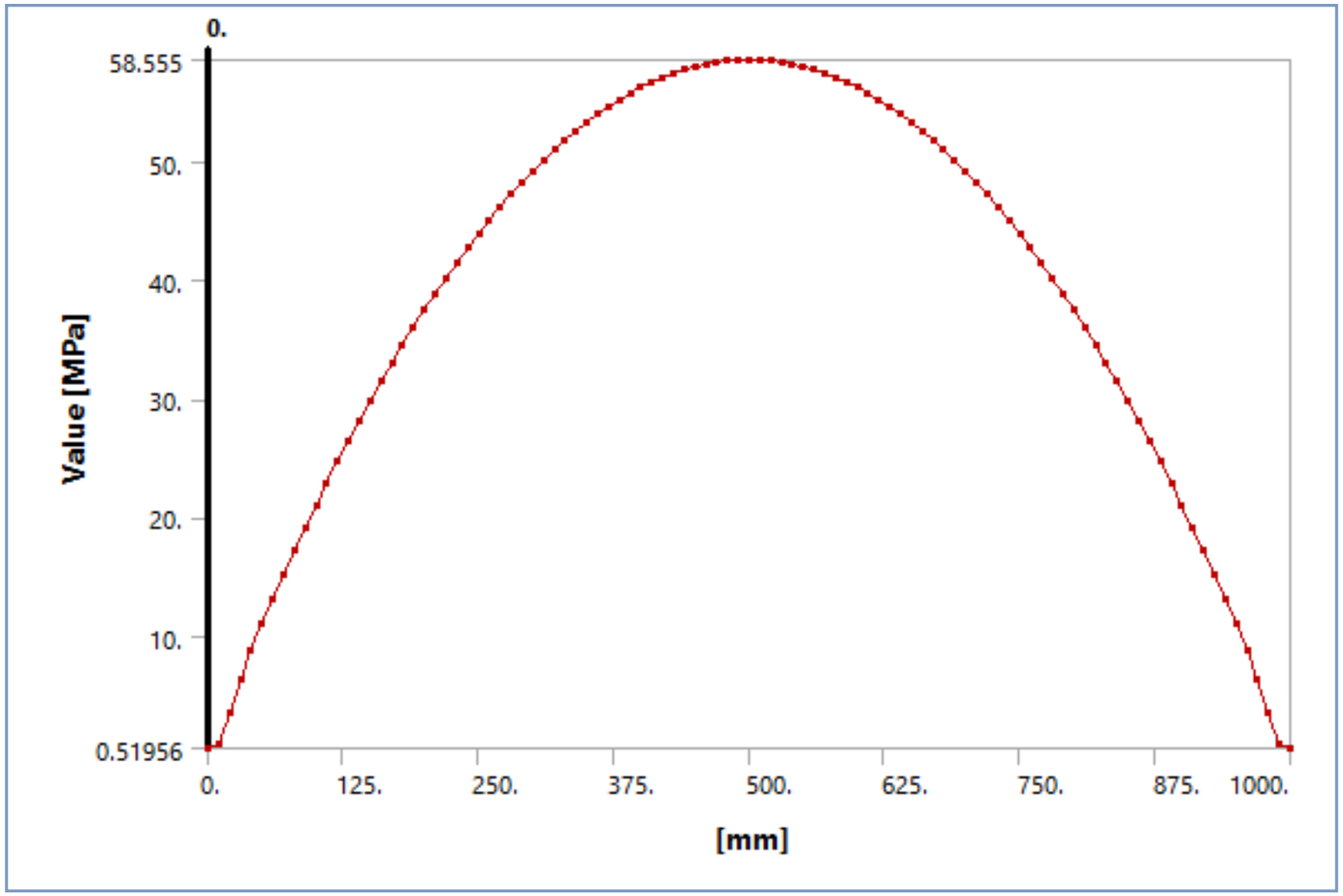


TABLE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

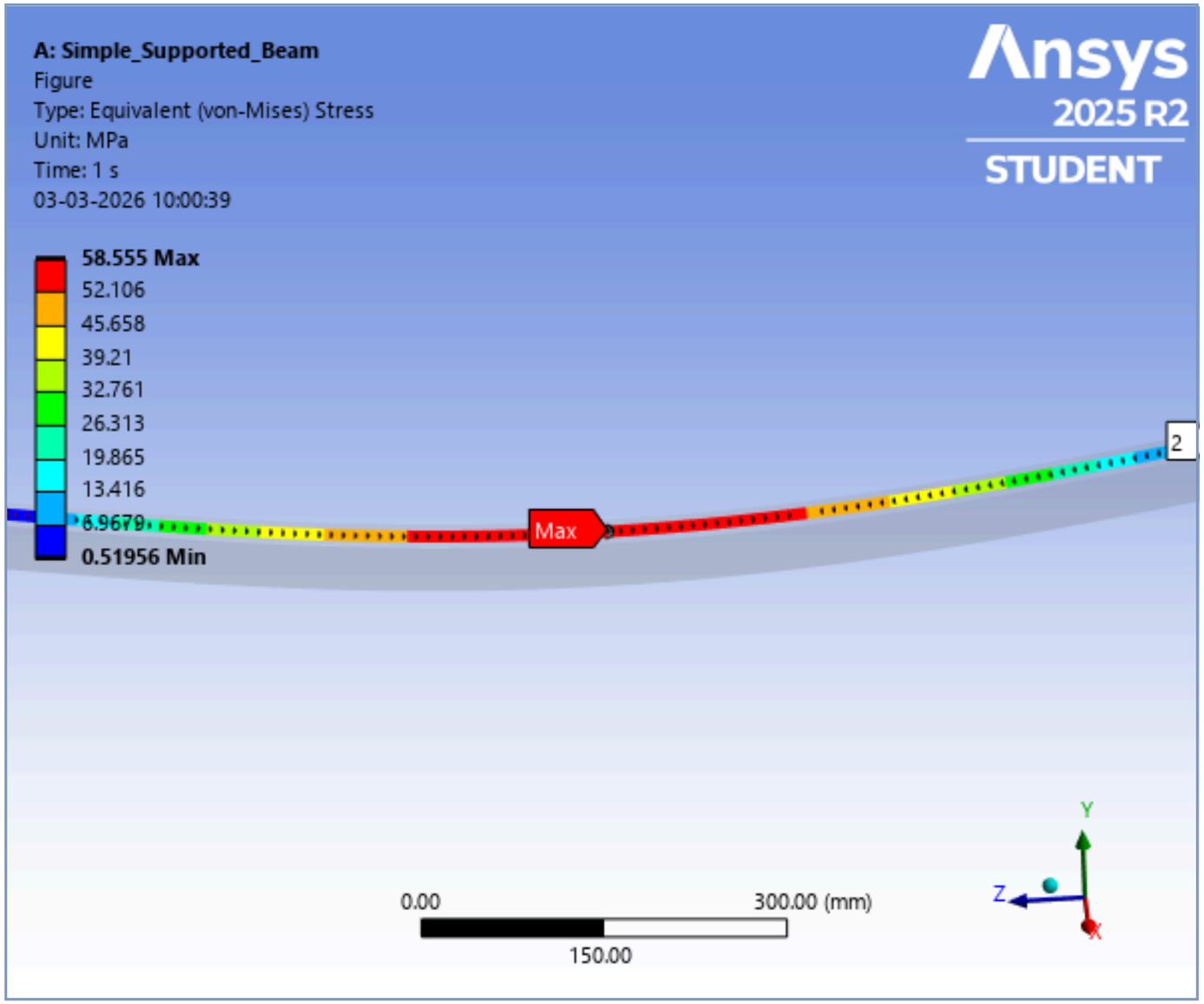
Length [mm]	Value [MPa]
0.	0.51956
10.	0.97076
20.	3.4405
30.	6.2896
40.	8.818
50.	11.076
60.	13.189
70.	15.228
80.	17.215
90.	19.158
100.	21.055
110.	22.906
120.	24.711

130.	26.469
140.	28.18
150.	29.844
160.	31.461
170.	33.031
180.	34.555
190.	36.031
200.	37.461
210.	38.844
220.	40.18
230.	41.469
240.	42.711
250.	43.906
260.	45.055
270.	46.156
280.	47.211
290.	48.219
300.	49.18
310.	50.094
320.	50.961
330.	51.781
340.	52.555
350.	53.281
360.	53.961
370.	54.594
380.	55.18
390.	55.719
400.	56.211
410.	56.656
420.	57.055
430.	57.406
440.	57.711
450.	57.969
460.	58.18
470.	58.344
480.	58.461
490.	58.531
500.	58.555
510.	58.531
520.	58.461
530.	58.344

540.	58.18
550.	57.969
560.	57.711
570.	57.406
580.	57.055
590.	56.656
600.	56.211
610.	55.719
620.	55.18
630.	54.594
640.	53.961
650.	53.281
660.	52.555
670.	51.781
680.	50.961
690.	50.094
700.	49.18
710.	48.219
720.	47.211
730.	46.156
740.	45.055
750.	43.906
760.	42.711
770.	41.469
780.	40.18
790.	38.844
800.	37.461
810.	36.031
820.	34.555
830.	33.031
840.	31.461
850.	29.844
860.	28.18
870.	26.469
880.	24.711
890.	22.906
900.	21.055
910.	19.158
920.	17.215
930.	15.228
940.	13.189

950.	11.076
960.	8.8181
970.	6.2894
980.	3.44
990.	0.97032
1000.	0.52001

FIGURE 7
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Figure



Material Data

Structural Steel

TABLE 19
Structural Steel > Constants

Density	7.85e-006 kg mm^-3
Coefficient of Thermal Expansion	1.2e-005 K^-1
Specific Heat	4.34e+005 mJ kg^-1 K^-1
Thermal Conductivity	6.05e-002 W mm^-1 K^-1
Resistivity	1.7e-004 ohm mm

TABLE 20
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 21
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 22
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 23
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 24
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 25
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295.15

TABLE 26
Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0
2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 27
Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 28
Structural Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
2.1e+005	0.3	1.75e+005	80769	

TABLE 29
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000